LESSON PLAN

NDT (19)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	09
Total No. of Lesson Plans	9	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Acoustic emission, basic concept, types, application, advantages
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Acoustic emission, basic concept, types, application, advantages
5. Specific Objectives	19.1.5.1 Trainee will be able to explain the optical holography and acoustic emission inspection

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of AE sensor placement on test coupon; Safety: Ensure no loose tools; Supervision: Instructor monitors sensor coupling
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of Non-Destructive Testing?
(b) List three common NDT methods used in aircraft maintenance.
(c) Define the term 'Acoustic Emission' in one sentence.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Basic concept of acoustic emission • Generation of elastic waves in materials • Types of acoustic emission sources • Detection and signal processing basics	L & Ds	PPT, IB	15 min
(b)	• Applications in aircraft structural health • Advantages over other NDT methods • Limitations and challenges in AE inspection • Comparison with optical holography	L & Ds	PPT, PDF	20 min
(c)	• Sensor coupling techniques • Data interpretation fundamentals • Standard reporting formats for AE	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how acoustic emission signals indicate structural damage.
(b) List two key advantages of AE compared to radiographic testing.

12. Home Assignment and Follow Up	Prepare a brief note on how AE inspection complements optical holography for future composite material inspections (Bridge to LP 02).
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	End of subject	Refer to the master syllabus reference list	-

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.
6. Clarify specific AE threshold values for aircraft aluminum alloys.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min